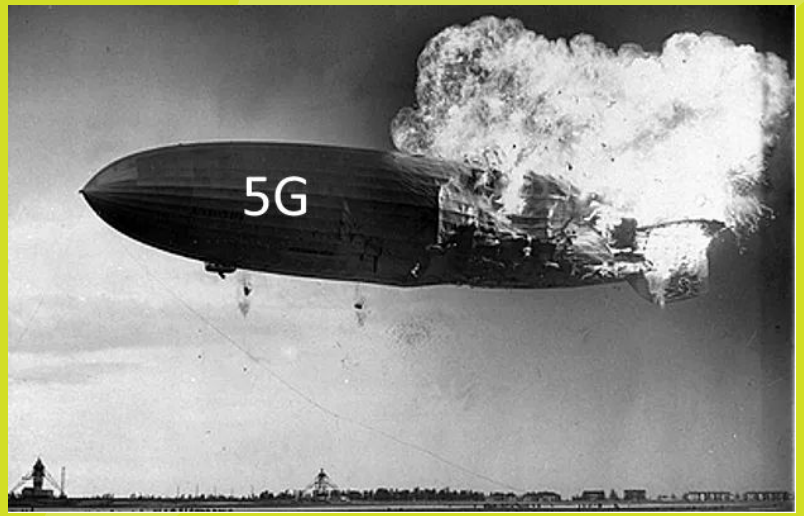


Preliminary Design Review “Service of 5G UHAA”

Skoltech

Skolkovo Institute of Science and Technology

#1

Responsibilities

Danil → System Architecture OPD, Prototyping plan

Valentin → System requirements, System Architecture SysML

Selamawit → Risk analysis

Haris → System model (IDEF0)

Artem → Schedule with WBS

Liaisan → Stakeholder expectation, Stakeholder analysis

Luis → Mission objective, team structure, ConOps

Mission Objectives

The mission objective is **to design the structural components and layout of the service UHAA**, including sufficient propulsion for steering, altitude controls, communications, and remote sensing.
The team should:

- Specify materials
- dimensions
- basic systems
- Create a scaled-down model (virtual + physical)
- prototype of the UHAA (virtual + physical)
- Collaborate with one of the Towing UHAA teams to design the connecting mechanism

Team Structure

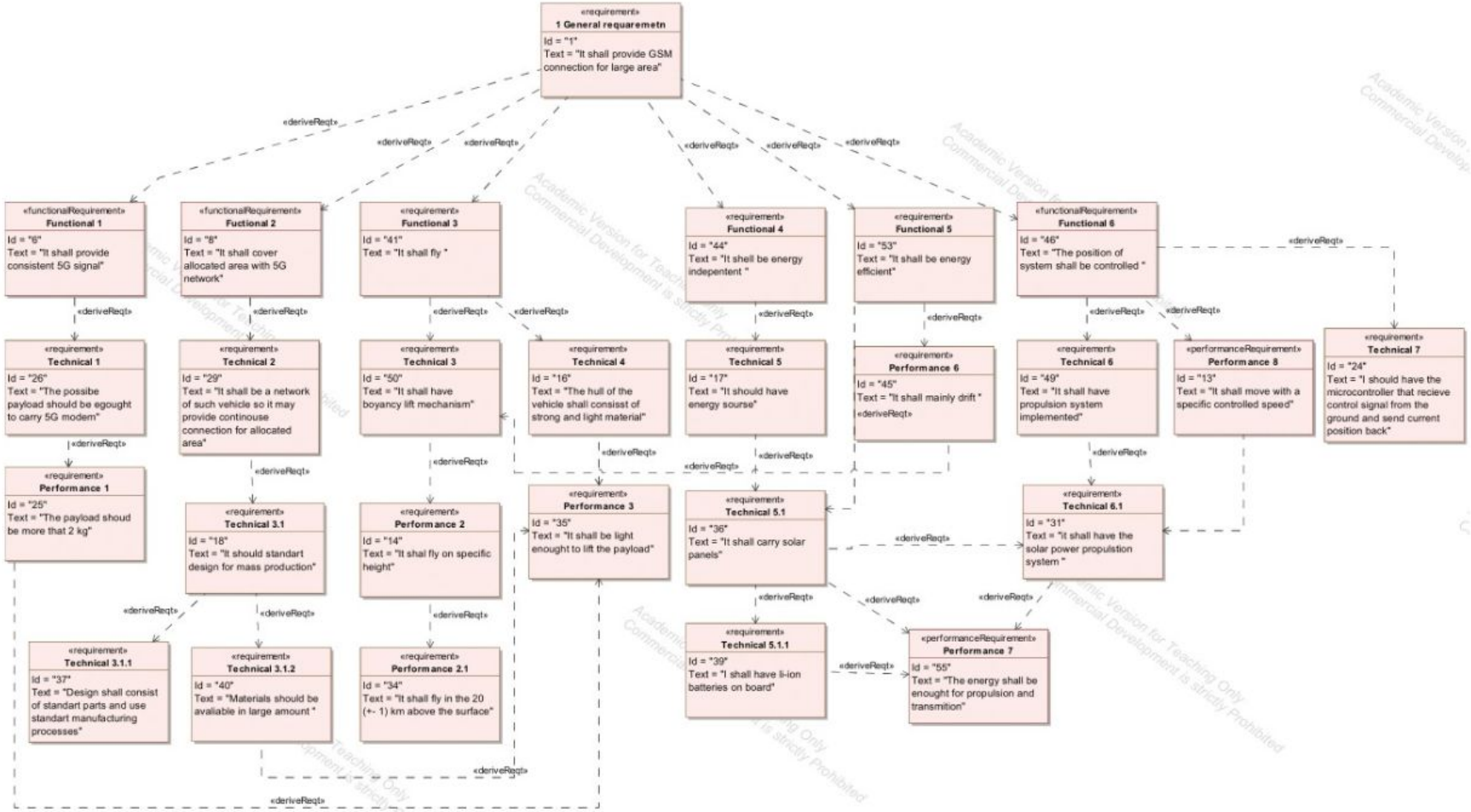
Team Leader : Valentin Tikhonenko
MBSE modeler : Luis Moreno
Analyst : Liaisan Safarova
Configuration and data manager : Selamawit Asfaw
Prototyping: Artyom Myshlyaev, Daniel Belov, Haris Khan

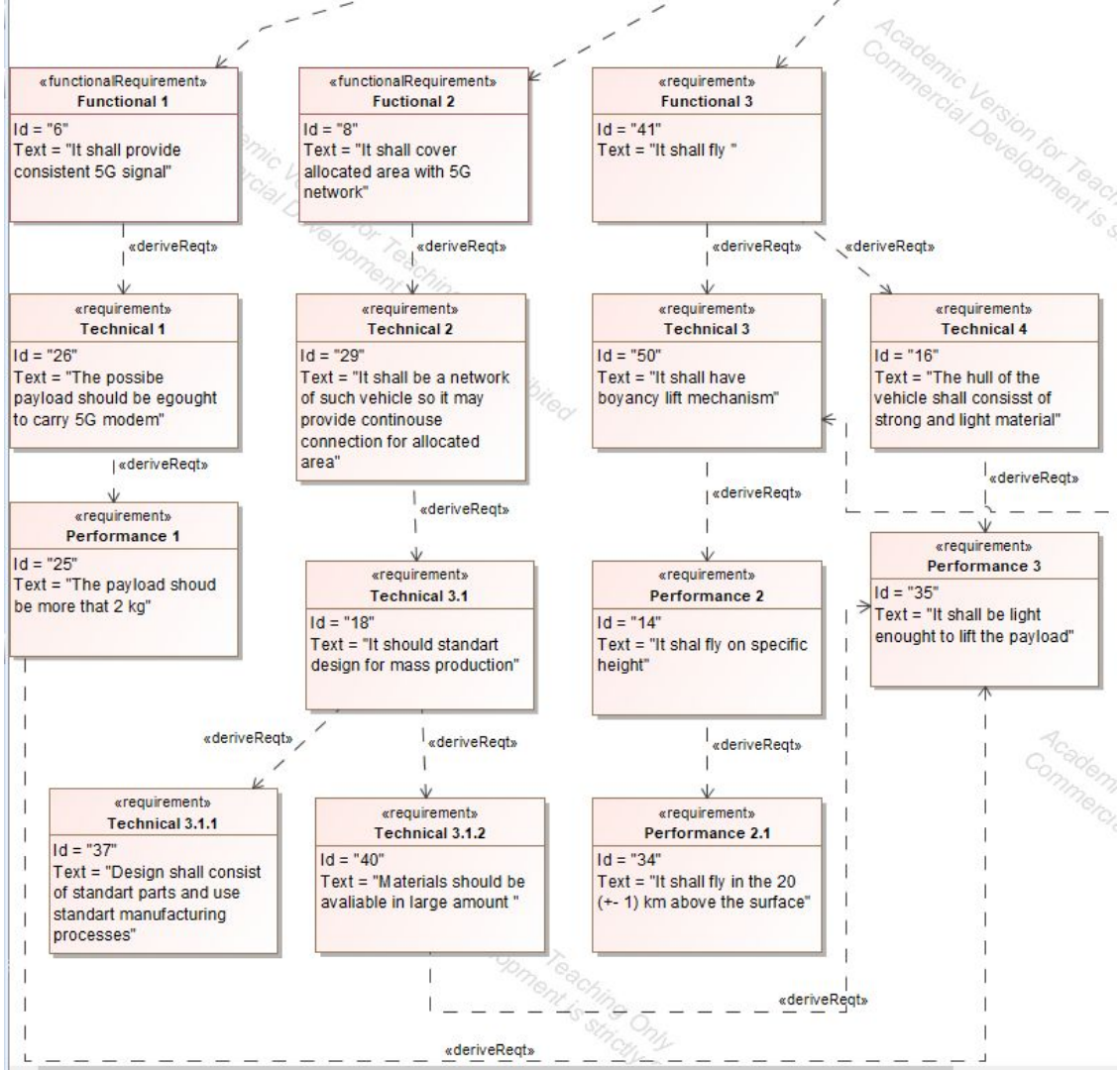
TEAM

#1

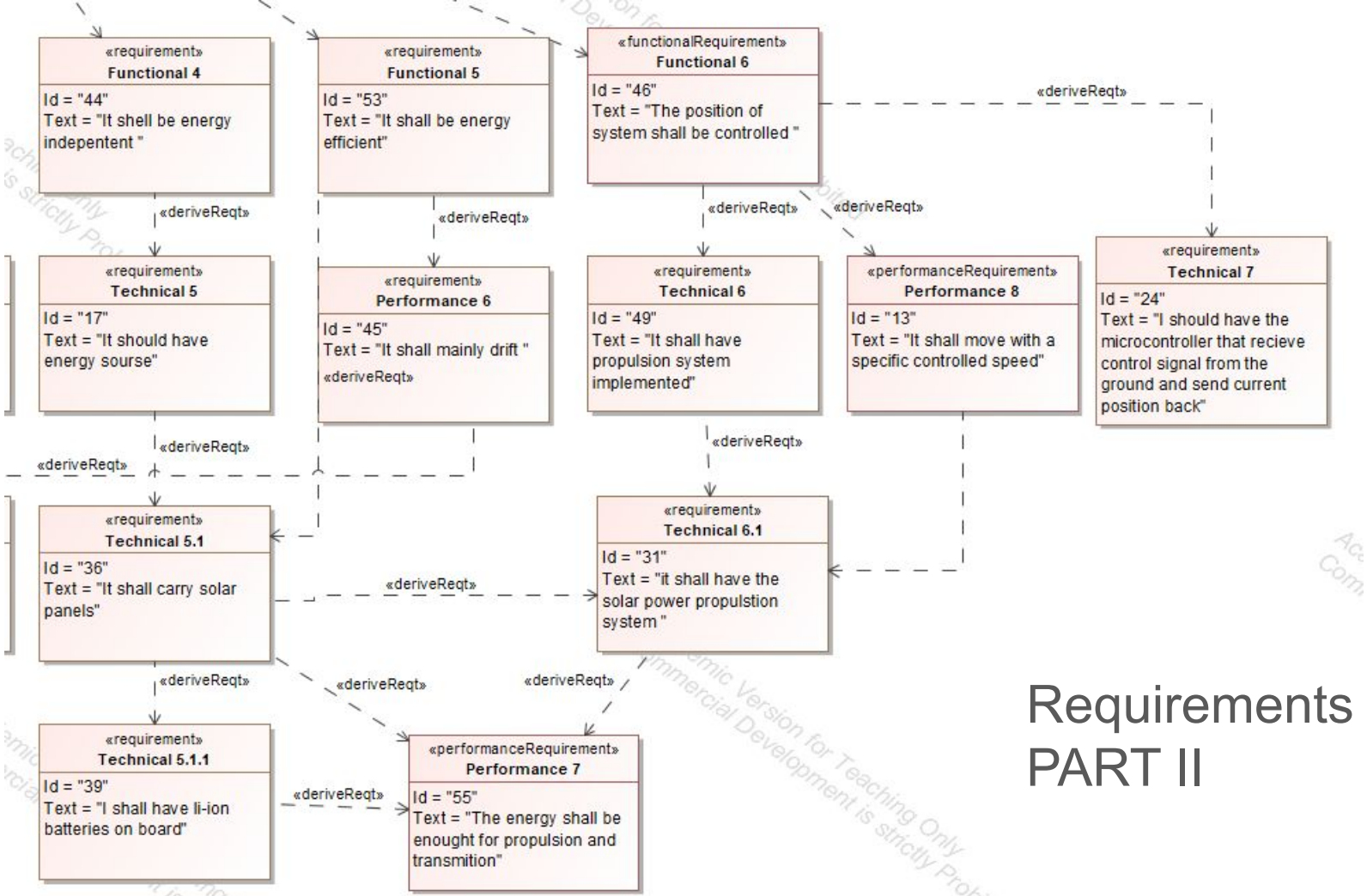
Conops


1 flying UMA at 20 +/- km
2 Drift using the wind
3 create mongnetwork of mobile base stations
4 Create close loop


«requirement»
1 General requaremetn
Id = "1"
Text = "It shall provide GSM connection for large area"
«functionalRequirement»
Functional 1
Id = "6"
Text = "It shall provide consistent 5G signal"
«functionalRequirement»
Fuctional 2
Id = "8"
Text = "It shall cover allocated area with 5G network"
«requirement»
Functional 3
Id = "41"
Text = "It shall fly "
«requirement»
Functional 4
Id = "44"
Text = "It shell be energy indepentent "
«requirement»
Functional 5
Id = "53"
Text = "It shall be energy efficient"
«functionalRequirement»
Functional 6
Id = "46"
Text = "The position of system shall be controlled "
«requirement»
Technical 1
Id = "26"
Text = "The possibe payload should be egought to carry 5G modem"
«requirement»
Technical 2
Id = "29"
Text = "It shall be a network of such vehicle so it may provide continouse connection for allocated area"
«requirement»
Technical 3
Id = "50"
Text = "It shall have boyancy lift mechanism"
«requirement»
Technical 4
Id = "16"
Text = "The hull of the vehicle shall consisst of strong and light material"
«requirement»
Technical 5
Id = "17"
Text = "It should have energy sourse"
«requirement»
Performance 6
Id = "45"
Text = "It shall mainly drift "
«requirement»
Technical 6
Id = "49"
Text = "It shall have propulsion system implemented"
«performanceRequirement»
Performance 8
Id = "13"
Text = "It shall move with a specific controlled speed"
«requirement»
Technical 7
Id = "24"
Text = "I should have the microcontroller that recieve control signal from the ground and send current position back"
«requirement»
Performance 1
Id = "25"
Text = "The payload shoud be more that 2 kg"
«requirement»
Technical 3.1
Id = "18"
Text = "It should standart design for mass production"
«requirement»
Performance 2
Id = "14"
Text = "It shal fly on specific height"
«requirement»
Performance 3
Id = "35"
Text = "It shall be light enought to lift the payload"
«requirement»
Technical 5.1
Id = "36"
Text = "It shall carry solar panels"
«requirement»
Technical 6.1
Id = "31"
Text = "it shall have the solar power propulstion system "
«requirement»
Technical 3.1.1
Id = "37"
Text = "Design shall consist of standart parts and use standart manufacturing processes"
«requirement»
Technical 3.1.2
Id = "40"
Text = "Materials should be avaliable in large amount "
«requirement»
Performance 2.1
Id = "34"
Text = "It shall fly in the 20 (+- 1) km above the surface"
«requirement»
Technical 5.1.1
Id = "39"
Text = "I shall have li-ion batteries on board"
«performanceRequirement»
Performance 7
Id = "55"
Text = "The energy shall be enought for propulsion and transmition"
«deriveReqt»



Requirements PART I



Requirements PART II

Prototyping plan

Vehicle control test prototype

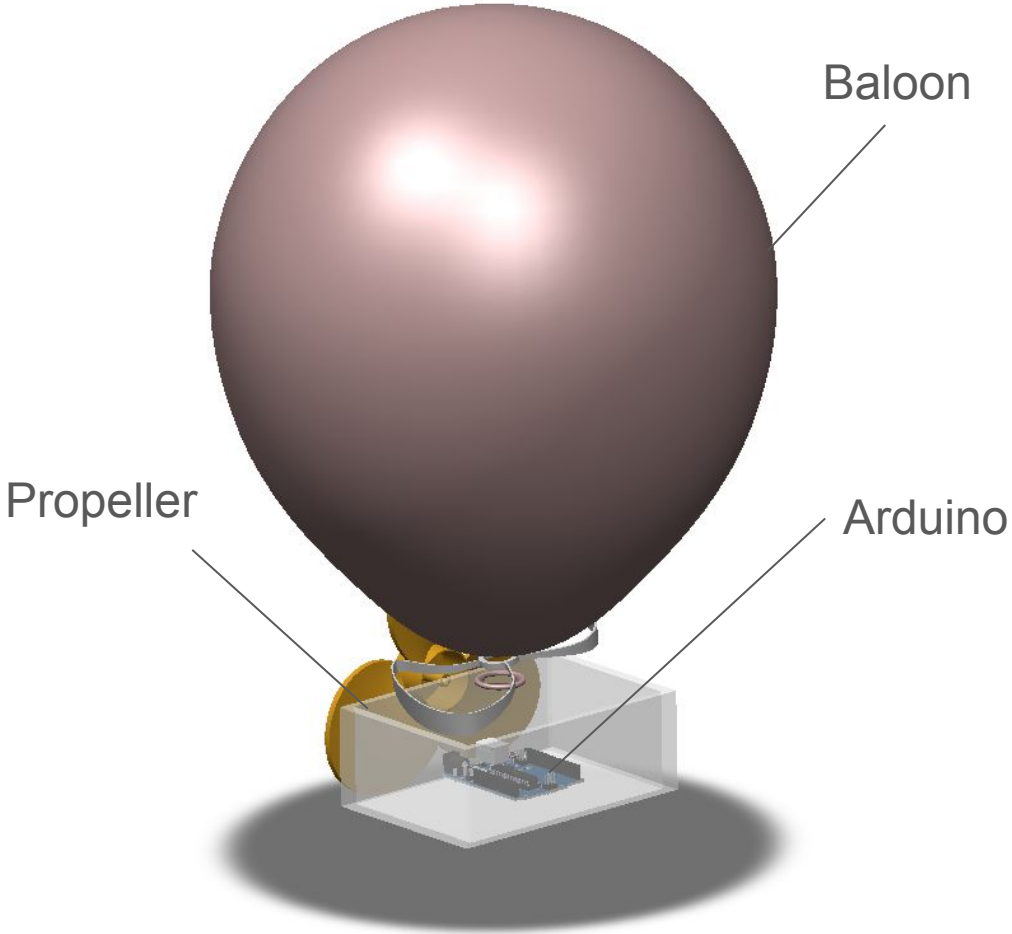
Zeppelin altitude test

Connection test

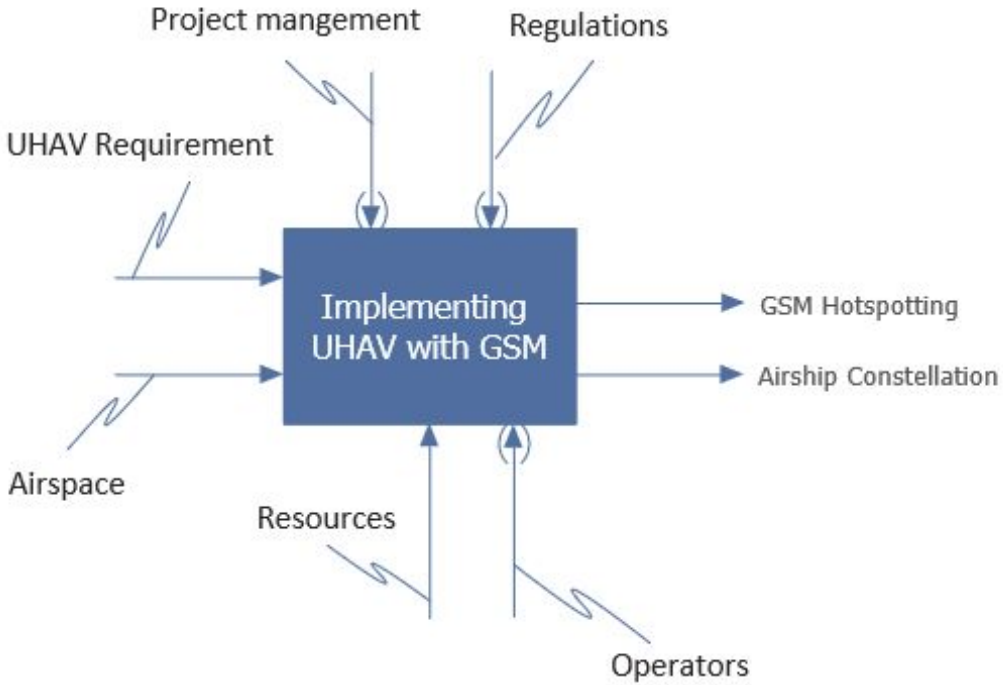
Network of vehicle control test

Solar panel power test

Coordinated uni

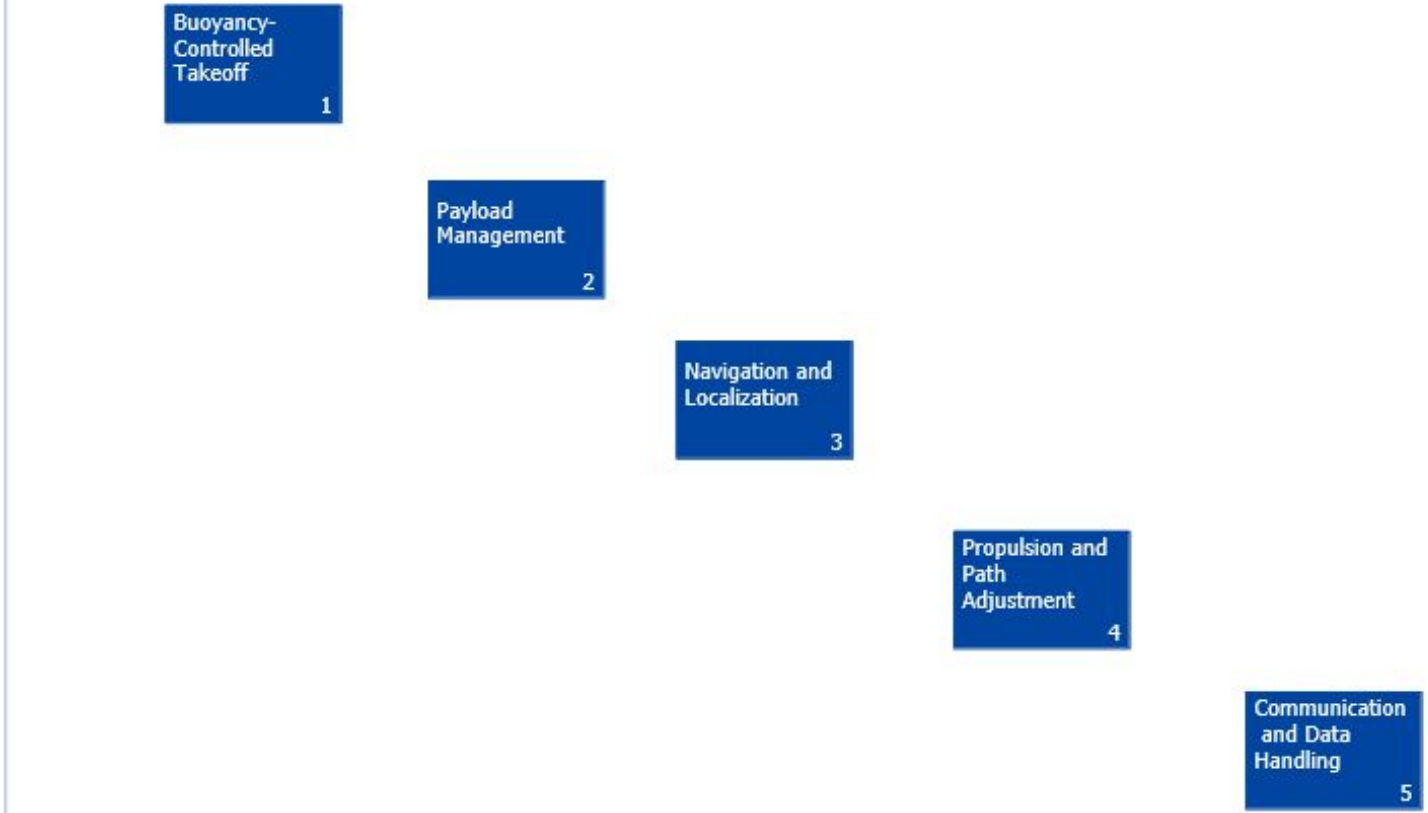


IDEFO

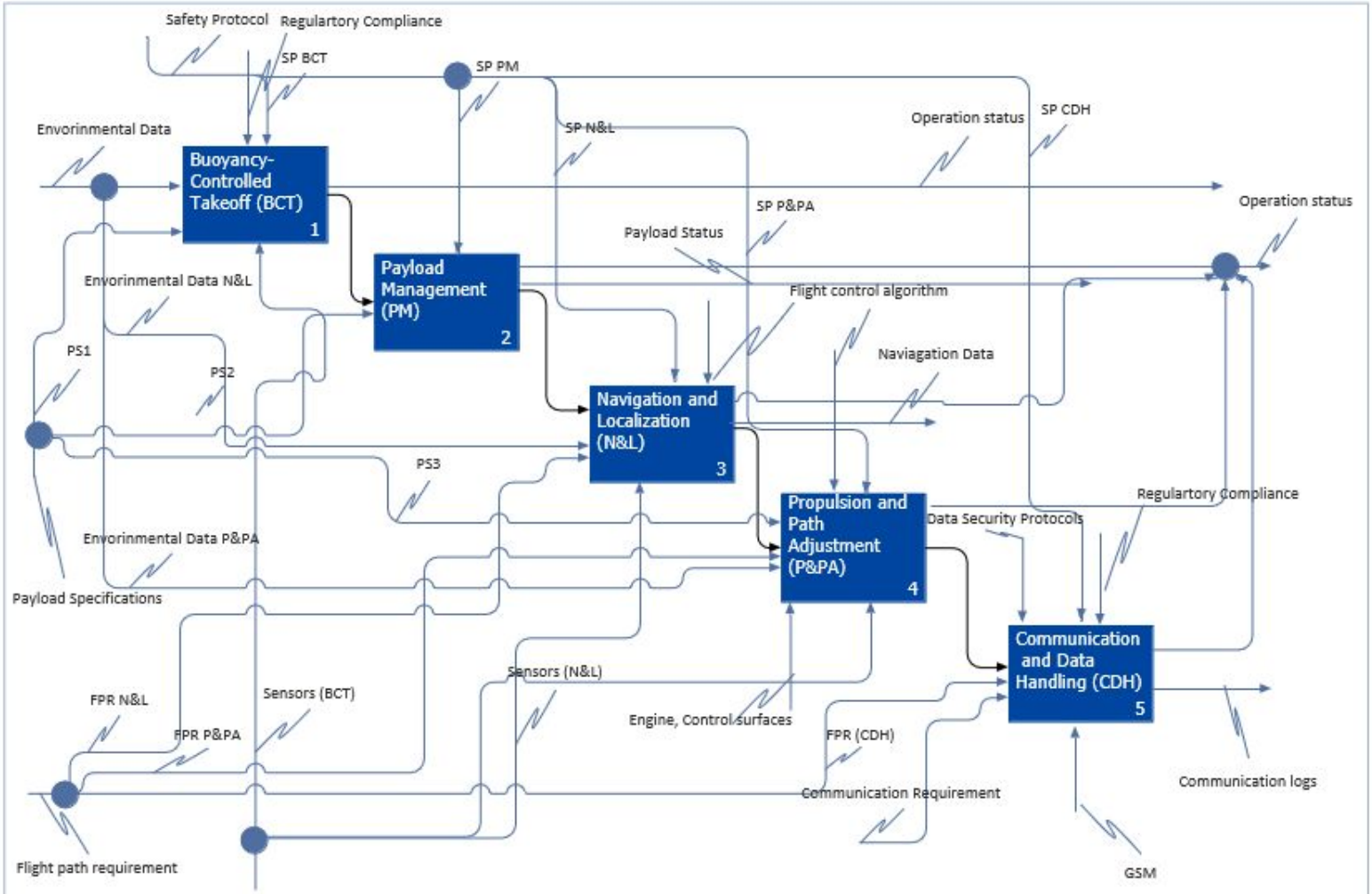


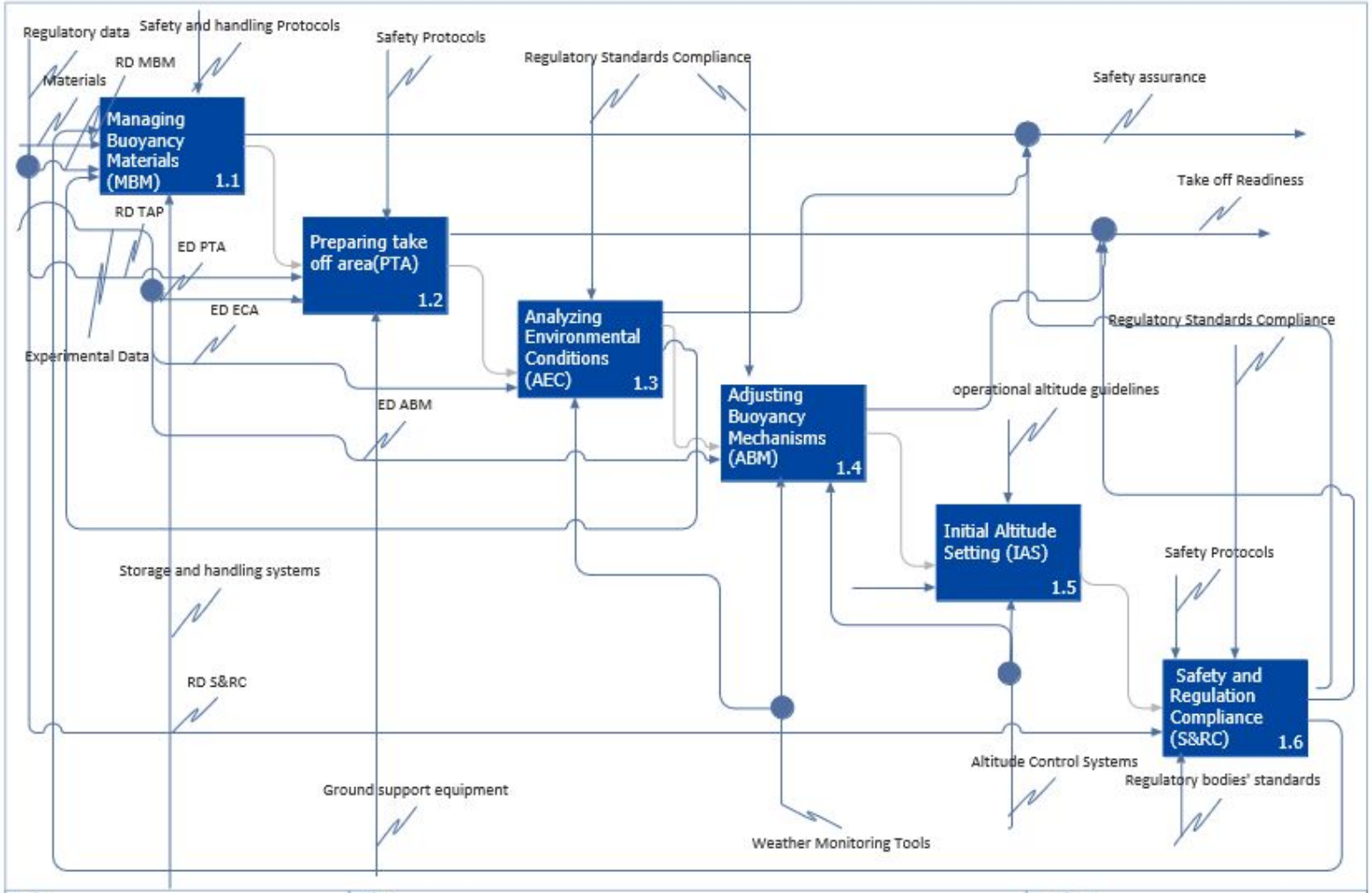
Purpose: Development of a more cost effective solution (Unmanned High Altitude Airship (UHAA)) perating at 20+/- km altitude as a platform

Viewpoint: Give an overview to the project manager for the implementation of UHAV

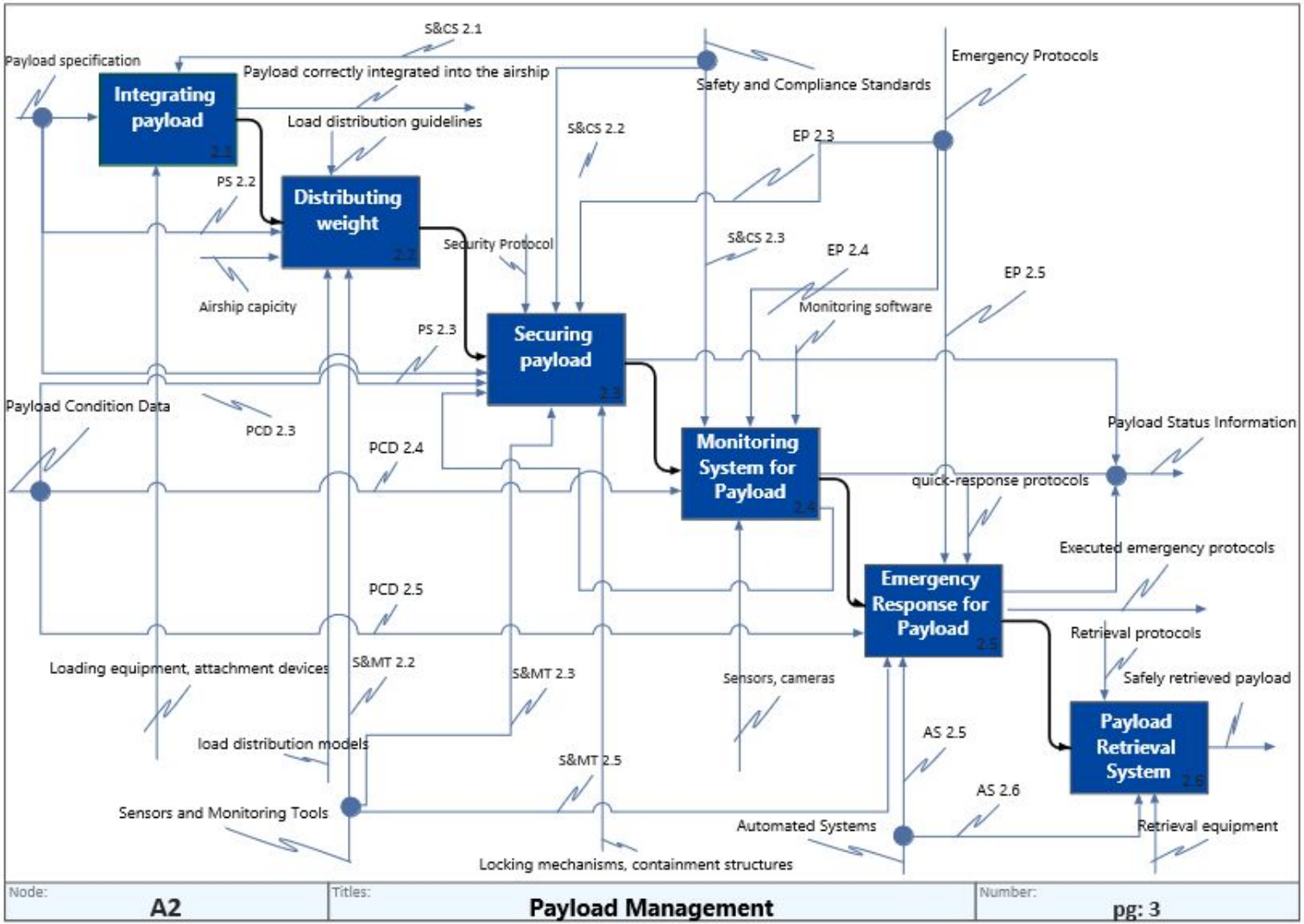


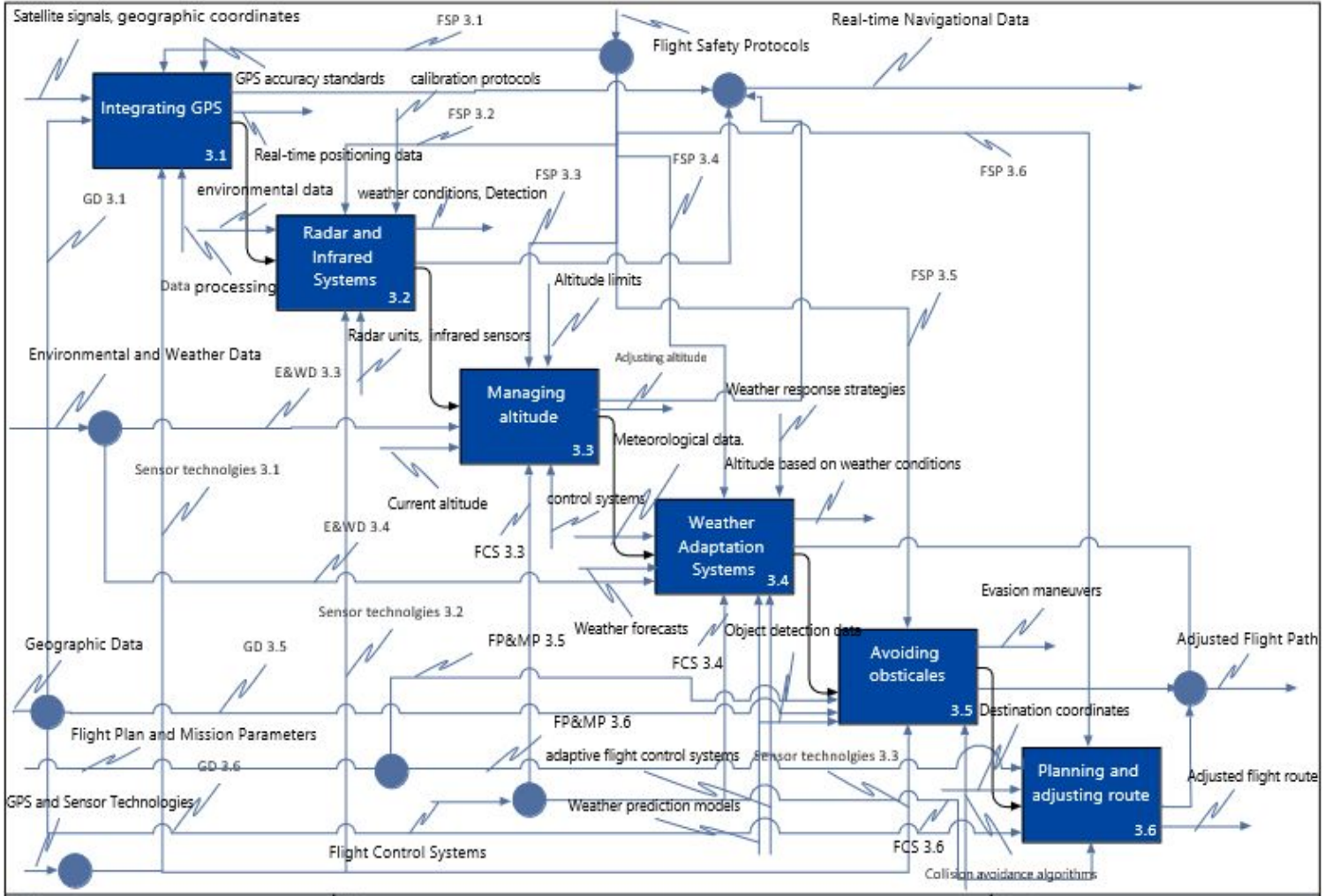
Node:	Titles:	Number:
AO	**Decomposition of the parent function "Implementing UHAV with GSM" to child funtion**	pg. 0

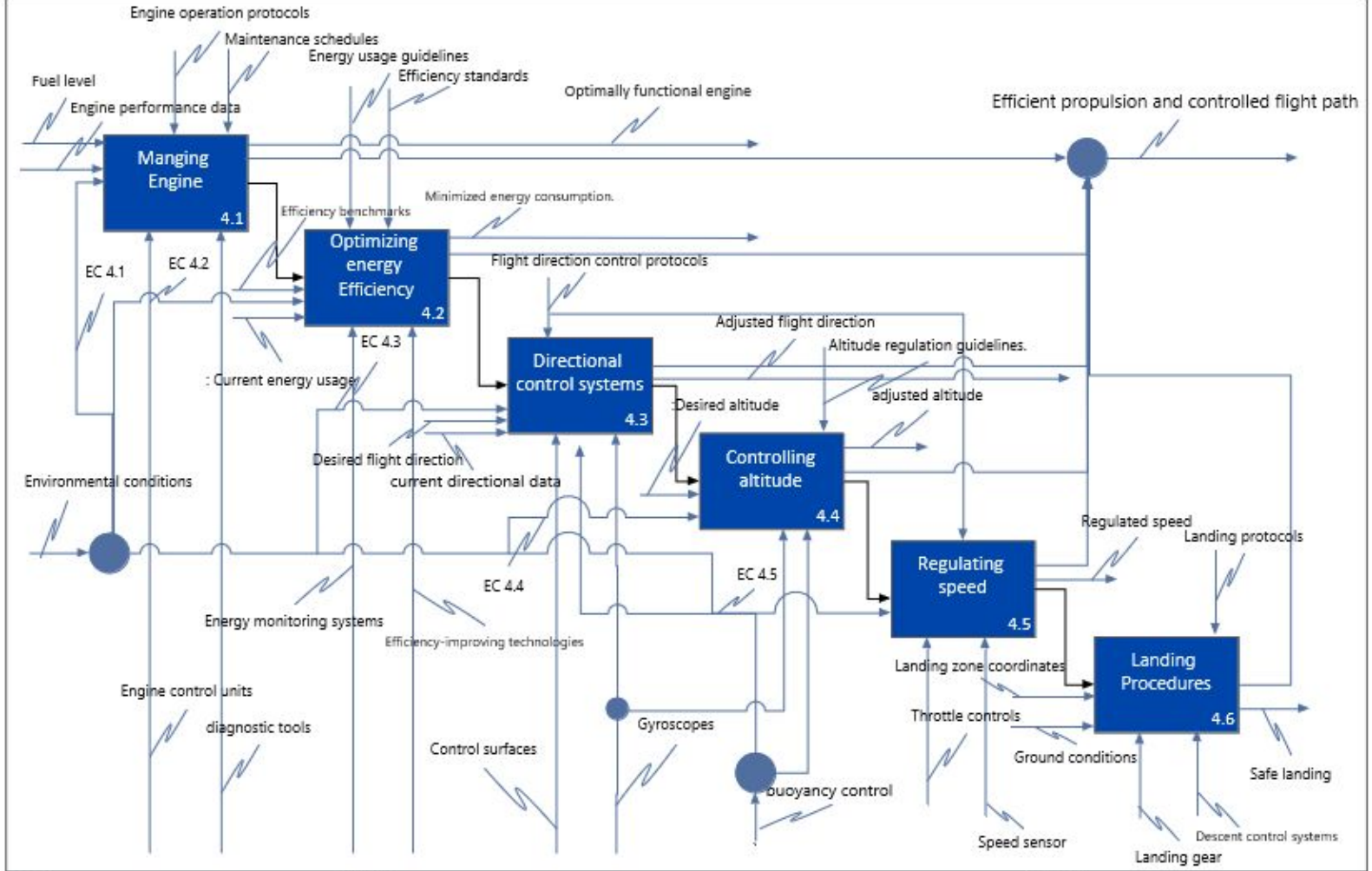

Safety Protocol
Regulartory Compliance
SP BCT
SP PM
Envorinmental Data
SP N&L
Operation status
SP CDH
Buoyancy-Controlled Takeoff (BCT) 1
SP P&PA
Payload Status
Operation status
Payload Management (PM) 2
Envorinmental Data N&L
Flight control algorithm
PS1
PS2
Naviagation Data
Navigation and Localization (N&L) 3
PS3
Regulartory Compliance
Propulsion and Path Adjustment (P&PA) 4
Data Security Protocols
Envorinmental Data P&PA
Payload Specifications
Communication and Data Handling (CDH) 5
Sensors (N&L)
FPR N&L
Sensors (BCT)
Engine, Control surfaces
FPR P&PA
FPR (CDH)
Communication logs
Communication Requirement
Flight path requirement
GSM
Node: A0
Titles: IDEFO: A0 representation of the child functions and relationship
Number: pg. 1



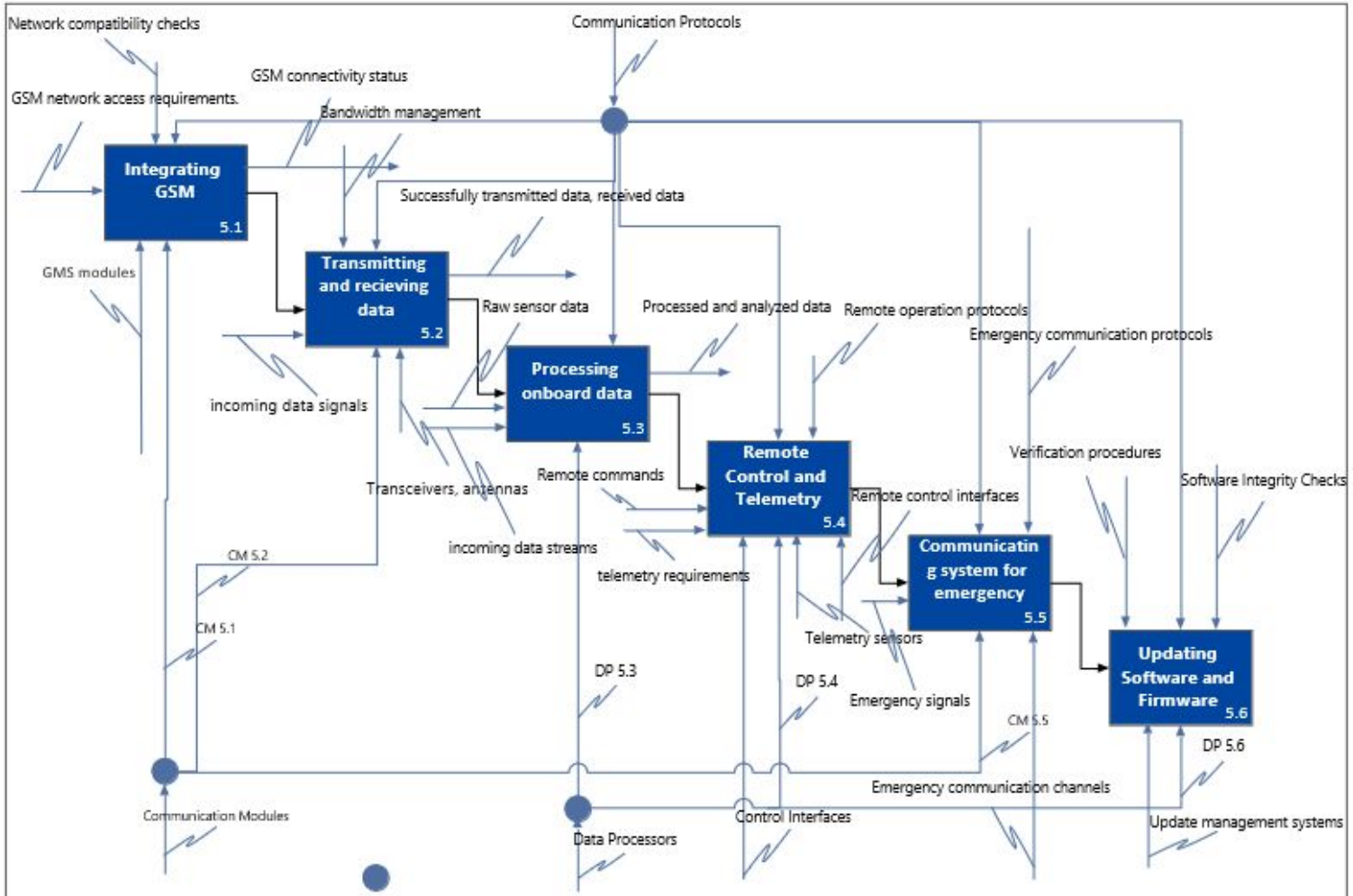
Node:	Titles:	Number:
A1	**Buoyancy-Controlled Takeoff**	pg. 2


Payload specification
S&CS 2.1
Payload correctly integrated into the airship
Integrating payload
2.1
Load distribution guidelines
Safety and Compliance Standards
Emergency Protocols
S&CS 2.2
EP 2.3
PS 2.2
Distributing weight
2.2
Security Protocol
S&CS 2.3
EP 2.4
EP 2.5
Airship capicity
Monitoring software
PS 2.3
Securing payload
2.3
Payload Condition Data
PCD 2.3
Payload Status Information
PCD 2.4
Monitoring System for Payload
2.4
quick-response protocols
Executed emergency protocols
PCD 2.5
Emergency Response for Payload
2.5
Retrieval protocols
Loading equipment, attachment devices
S&MT 2.2
S&MT 2.3
Sensors, cameras
Safely retrieved payload
load distribution models
AS 2.5
Payload Retrieval System
2.6
S&MT 2.5
AS 2.6
Sensors and Monitoring Tools
Automated Systems
Retrieval equipment
Locking mechanisms, containment structures
Node:
A2
Titles:
Payload Management
Number:
pg: 3


Satellite signals, geographic coordinates
FSP 3.1
Flight Safety Protocols
Real-time Navigational Data
Integrating GPS
3.1
GPS accuracy standards
calibration protocols
FSP 3.2
Real-time positioning data
FSP 3.4
FSP 3.6
FSP 3.3
environmental data
weather conditions, Detection
GD 3.1
Radar and Infrared Systems
3.2
Data processing
Altitude limits
FSP 3.5
Radar units, infrared sensors
Environmental and Weather Data
E&WD 3.3
Adjusting altitude
Managing altitude
3.3
Weather response strategies
Meteorological data.
Sensor technolgies 3.1
Altitude based on weather conditions
Current altitude
control systems
E&WD 3.4
FCS 3.3
Weather Adaptation Systems
3.4
Evasion maneuvers
Sensor technolgies 3.2
Geographic Data
GD 3.5
FP&MP 3.5
Weather forecasts
Object detection data
Adjusted Flight Path
FCS 3.4
Avoiding obsticales
3.5
Destination coordinates
Flight Plan and Mission Parameters
FP&MP 3.6
adaptive flight control systems
Sensor technolgies 3.3
GD 3.6
Planning and adjusting route
3.6
Adjusted flight route
GPS and Sensor Technologies
Weather prediction models
Flight Control Systems
FCS 3.6
Collision avoidance algorithms
Node:
A3
Titles:
Navigation and Localization
Number:
pg: 4

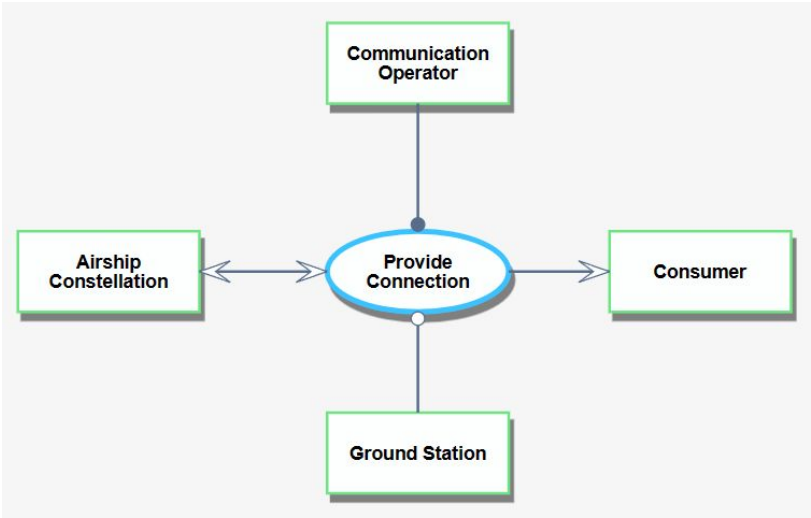


Node:	Titles:	Number:
A4	**Propulsion and Path Adjustment**	Pg: 5

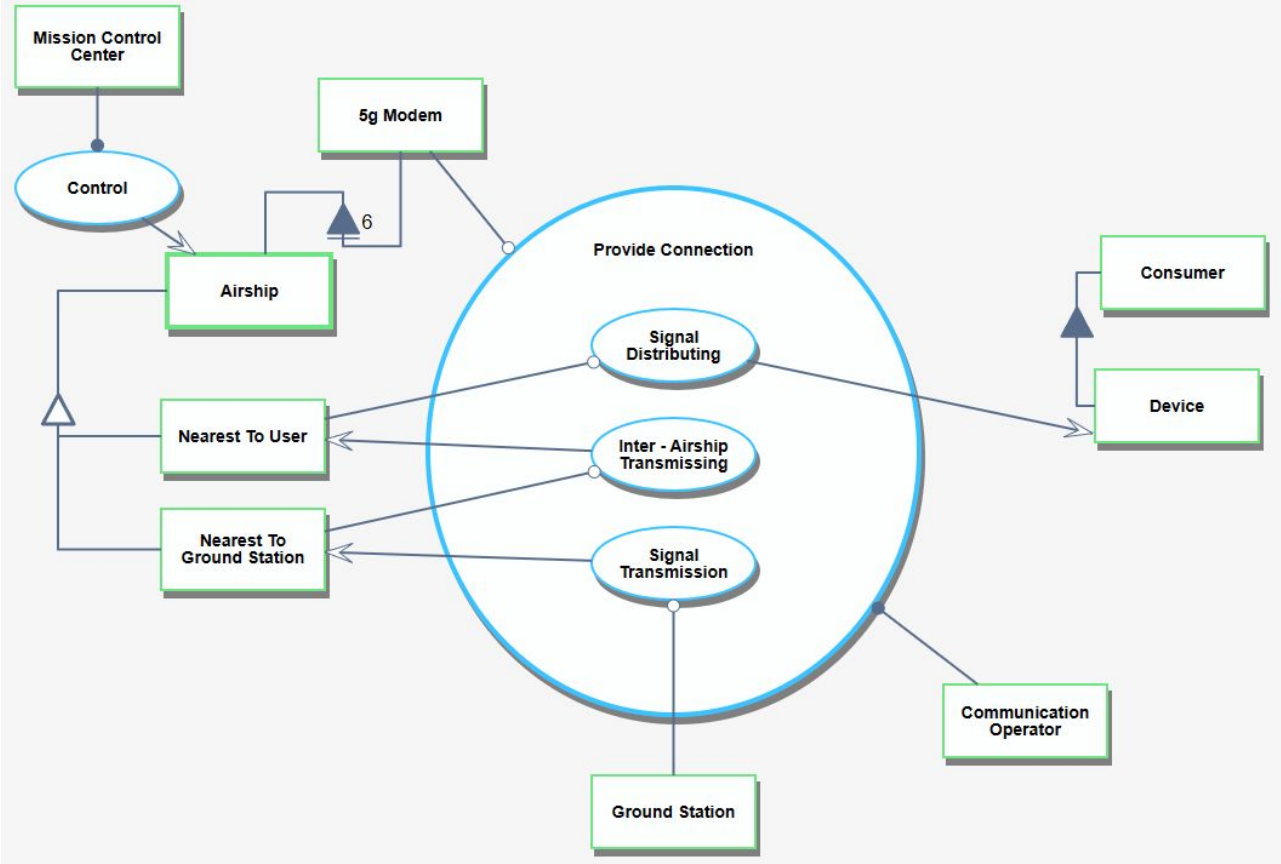


Node:	Titles:	Number:
A5	**Communication and Data Handling System**	pg: 6

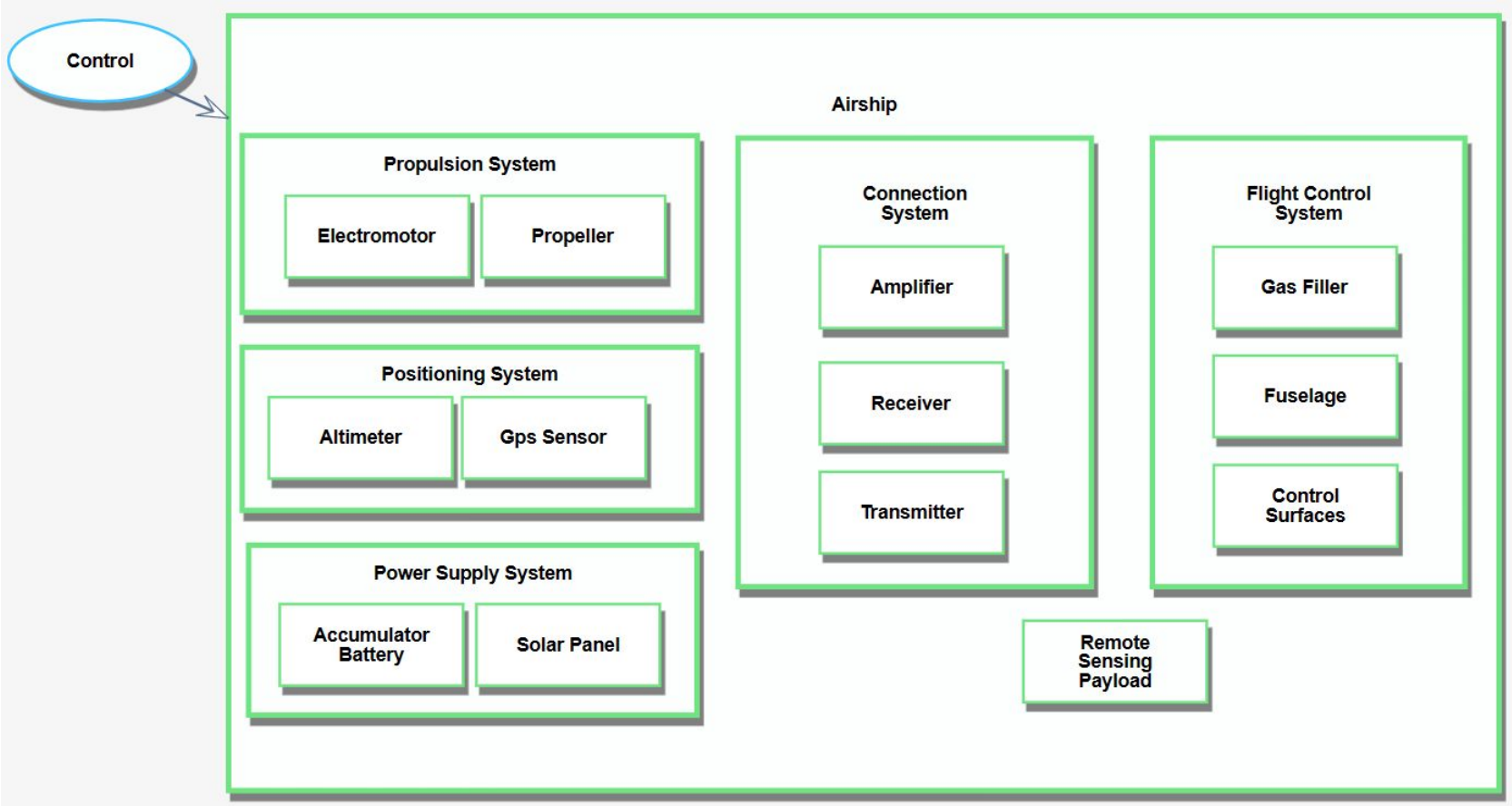
System Architecture



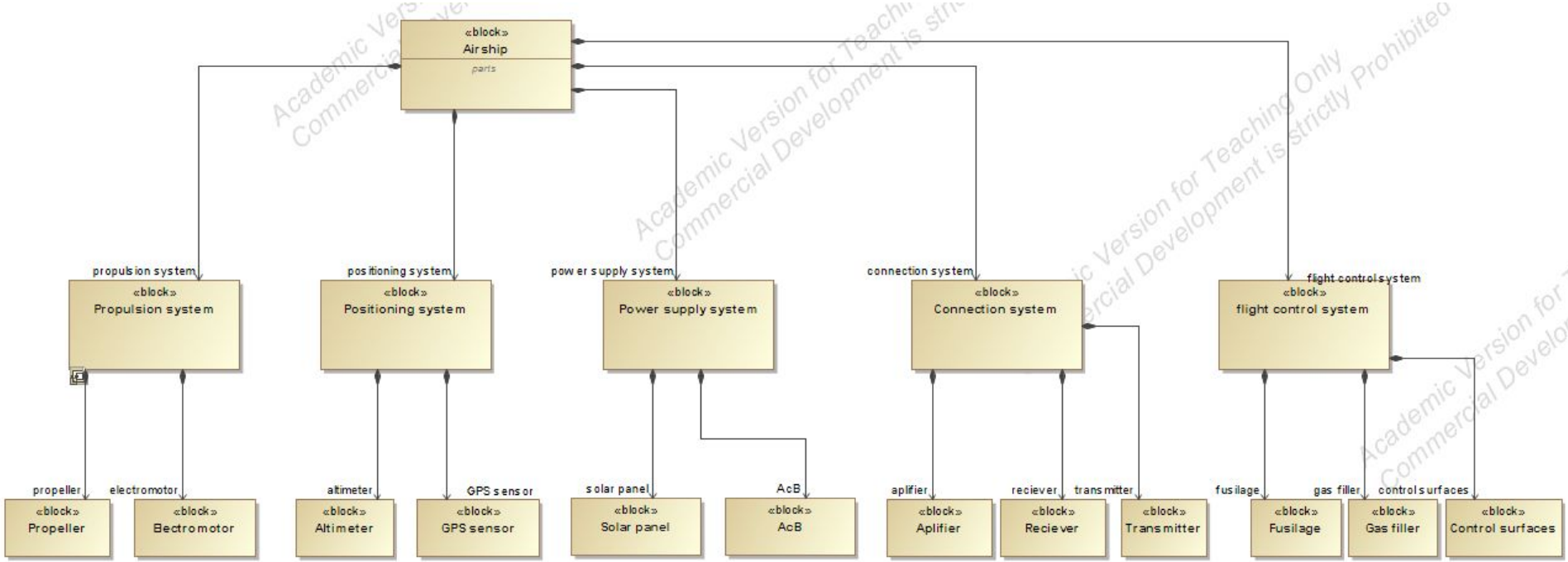
System Architecture



System Architecture



SysML System architecture



Risk analysis

01	Technological Challenges	• Advanced technologies and engineering solutions • Technical difficulties • Impact the aircraft's performance, reliability, or safety
02	Regulatory and Certification	• complex and time-consuming • delays or complications in obtaining approvals from aviation authorities • impact the development timeline and overall project cost.
03	Safety Concerns	• safety of passengers and crew, • payloads is paramount • exposure to cosmic radiation
04	Cost Overruns	• exceeding the budget due to unexpected expenses • design iterations, technological setbacks or changes in project scope • Poor cost management (feasibility and long-term sustainability)
05	Environmental Impact	• emissions and noise • require additional research, technology development, or operational constraints.

06	Market Viability	• insufficient market demand • competition from existing alternatives,
07	Supply Chain and Manufacturing Risks	• Developing and assembling a complex airship • delays, quality issues, or disruptions • Building strong relationships with suppliers
08	Intellectual Property Protection	• intellectual property infringement • unauthorized use • leakage of proprietary information
09	Integration and System Complexity	• subsystems and components • compatibility issues • system failures
10	Stakeholder Management	• Effective communication • misalignment of expectations, conflicts of interest • challenges in managing stakeholder relationships

Risk ID/Assessment

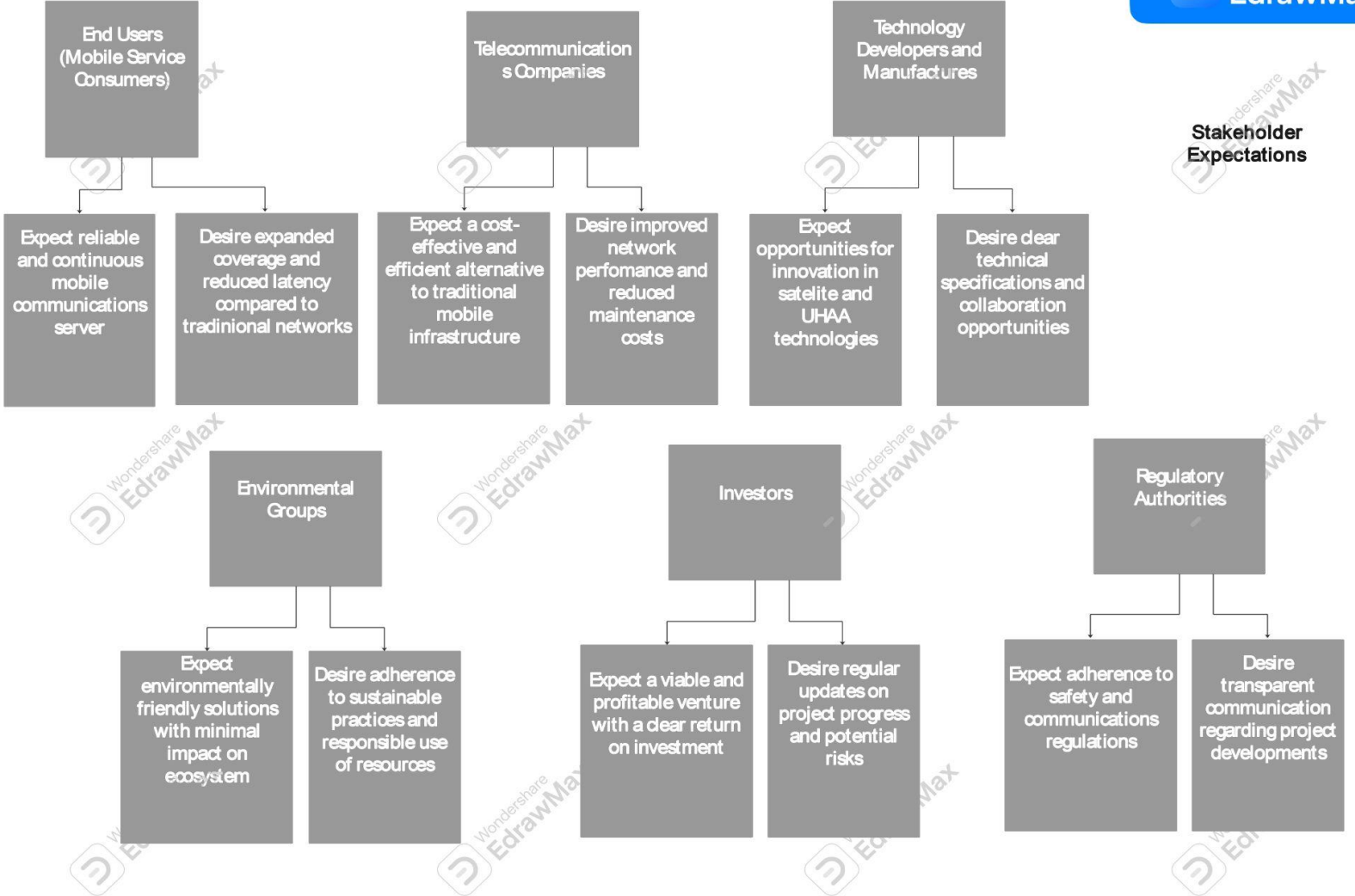
ID	Name	Probability	Impact
A	Technological Challenges	4	4
B	Regulatory and Certification Requirements	3	4
C	Safety Concerns	4	5
D	Cost Overruns	3	4

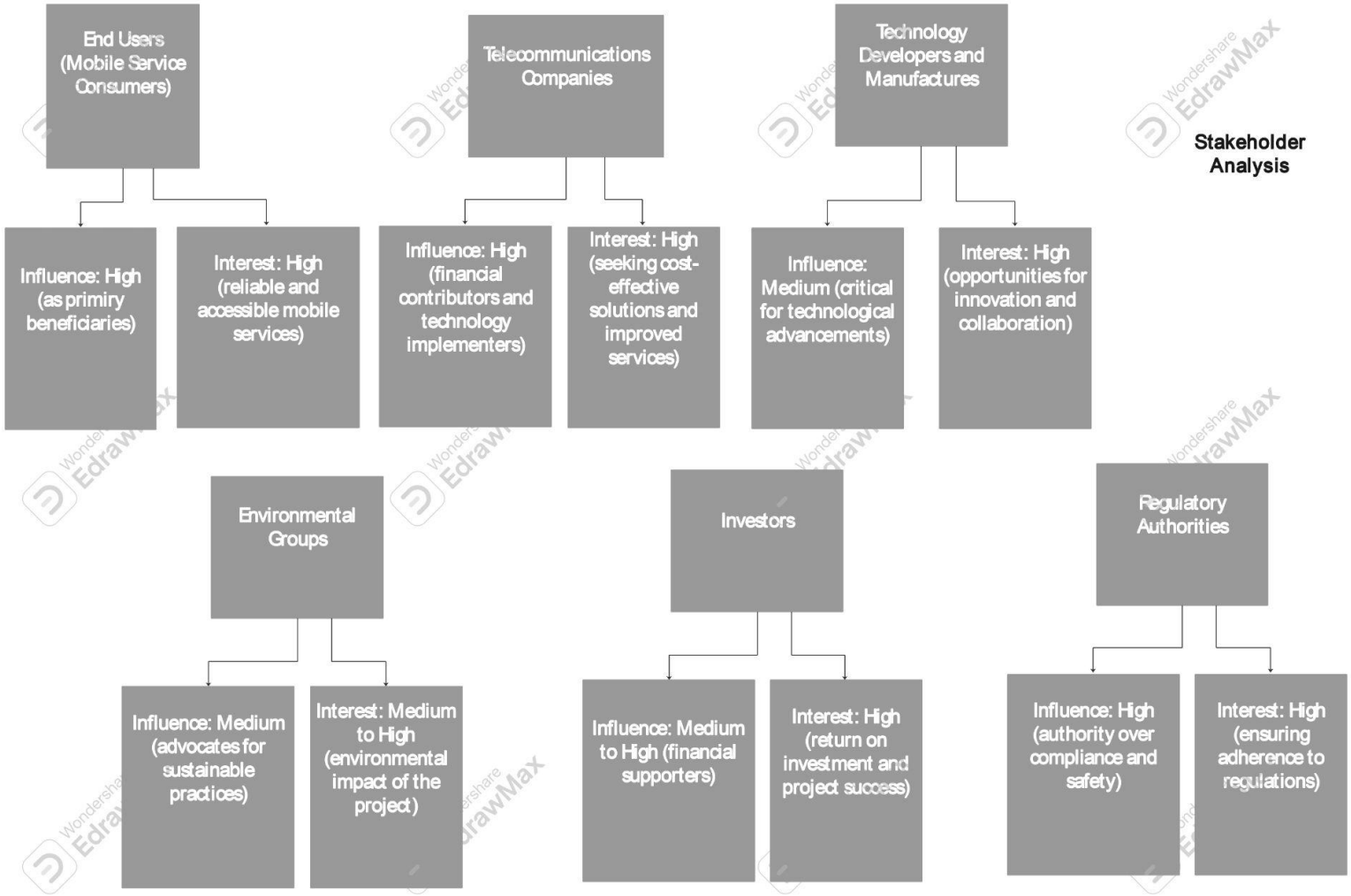
Risk ID/Assessment

ID	Name	Probability	Impact
E	Market Viability	3	4
F	Supply Chain and Manufacturing Risks	3	3
G	Intellectual Property Protection	3	4
H	Integration and System Complexity	4	4
I	Stakeholder Management	2	3

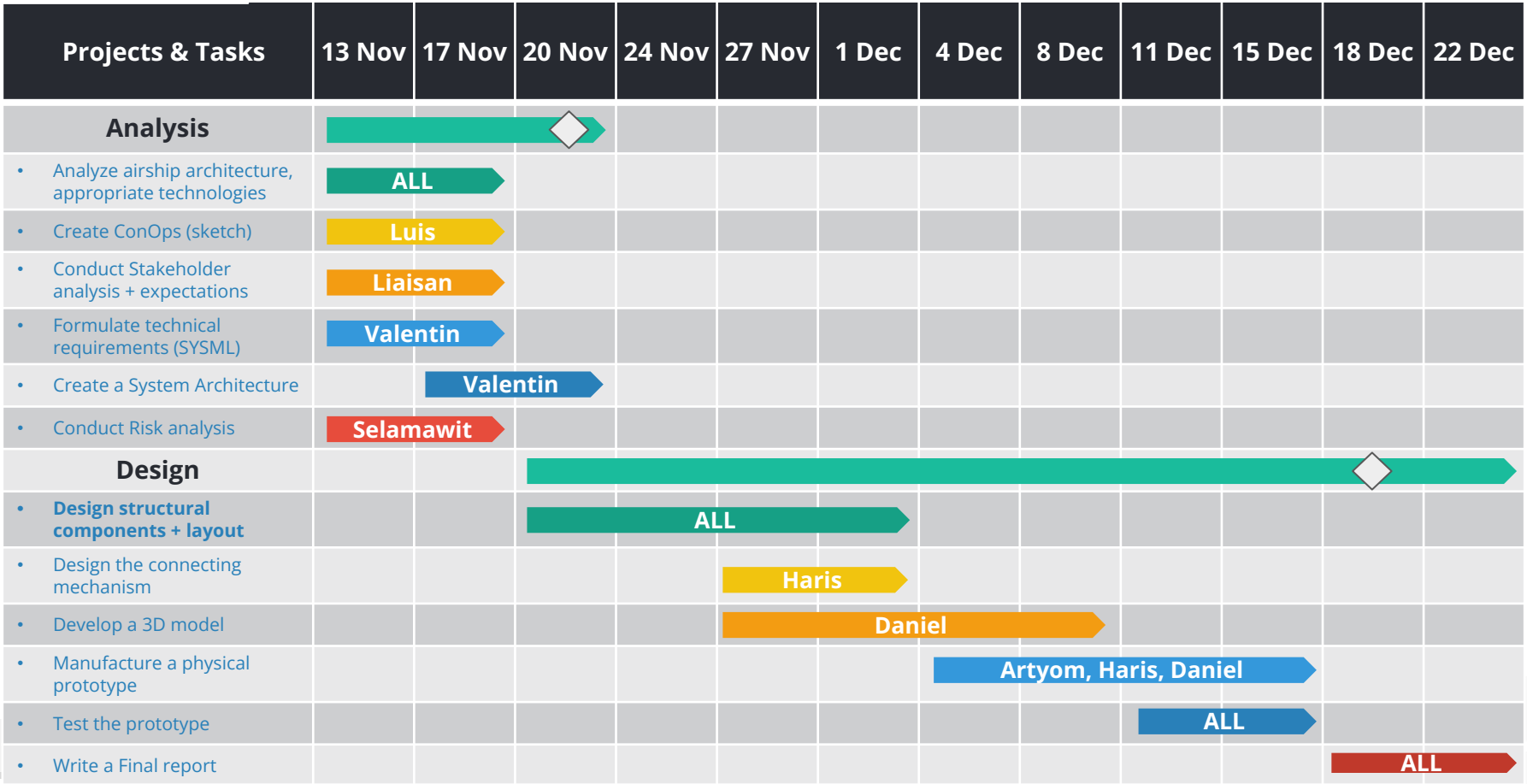
Risk Sector Plot

Probability/ Impact	1	2	3	4	5
1					
2			I		
3			F	B , D ,E, G	
4				A,H	C
5					


Stakeholder Expectations
End Users (Mobile Service Consumers)
Expect reliable and continuous mobile communications server
Desire expanded coverage and reduced latency compared to tradinional networks
Telecommunications Companies
Expect a cost-effective and efficient alternative to traditional mobile infrastructure
Desire improved network perfomance and reduced maintenance costs
Technology Developers and Manufactures
Expect opportunities for innovation in satelite and UHAA technologies
Desire clear technical specifications and collaboration opportunities
Environmental Groups
Expect environmentally friendly solutions with minimal impact on ecosystem
Desire adherence to sustainable practices and responsible use of resources
Investors
Expect a viable and profitable venture with a clear return on investment
Desire regular updates on project progress and potential risks
Regulatory Authorities
Expect adherence to safety and communications regulations
Desire transparent communication regarding project developments


End Users (Mobile Service Consumers)
Influence: High (as primiry beneficiaries)
Interest: High (reliable and accessible mobile services)
Telecommunications Companies
Influence: High (financial contributors and technology implementers)
Interest: High (seeking cost-effective solutions and improved services)
Technology Developers and Manufactures
Influence: Medium (critical for technological advancements)
Interest: High (opportunities for innovation and collaboration)
Stakeholder Analysis
Environmental Groups
Influence: Medium (advocates for sustainable practices)
Interest: Medium to High (environmental impact of the project)
Investors
Influence: Medium to High (financial supporters)
Interest: High (return on investment and project success)
Regulatory Authorities
Influence: High (authority over compliance and safety)
Interest: High (ensuring adherence to regulations)

Schedule


Projects & Tasks
13 Nov
17 Nov
20 Nov
24 Nov
27 Nov
1 Dec
4 Dec
8 Dec
11 Dec
15 Dec
18 Dec
22 Dec
Analysis
Analyze airship architecture, appropriate technologies
ALL
Create ConOps (sketch)
Luis
Conduct Stakeholder analysis + expectations
Liaisan
Formulate technical requirements (SYSML)
Valentin
Create a System Architecture
Valentin
Conduct Risk analysis
Selamawit
Design
Design structural components + layout
ALL
Design the connecting mechanism
Haris
Develop a 3D model
Daniel
Manufacture a physical prototype
Artyom, Haris, Daniel
Test the prototype
ALL
Write a Final report
ALL

Design WBS

1 Flight control system
    1.1 Develop control system in collaboration with control team
    1.2 Design Lift system
        1.1 Define lift principle
        1.2 Define math model
        1.3 Define structure parameters and constraints
    1.3 Design control surfaces
2 Propulsion system
    2.1 Define engine parameter
    2.2 Develop propeller
3. Positioning system
    3.1 Define the altimeter principle
    3.2 Place and connect GPS sensors
4. Power supply system
    4.1 Determine solar panel required surface and power need for operating
    4.2 Determine accumulator capacity
5. Connection system
    5.1 Design position and structure of the case for internet equipment